

## 2.1 Operation Contract: executeDelivery

**Operation:** executeDelivery(deliveryId : int, currentStep : int, recordedWeight : double)

**Cross References:** Use Cases: Execute Delivery

**Preconditions:**

1. A Delivery instance uniquely identified by deliveryId exists in the system memory.
2. The status attribute of the targeted Delivery instance is READY or EXECUTING.
3. The currentStep parameter matches the exact active step counter within the targeted Delivery instance.

**Postconditions:**

- If currentStep evaluated to 0:
  - The status attribute of the targeted Delivery instance was modified to EXECUTING (attribute modification).
- If recordedWeight is less than or equal to the maximum capacity weight of the associated Truck instance (Success Path):
  - The recordedWeight attribute within the targeted Delivery instance was updated to the value of the recordedWeight parameter (attribute modification).
  - If currentStep represents the final destination index of the route sequence: The status attribute of the targeted Delivery instance was modified to COMPLETED (attribute modification).
  - If currentStep is less than the final destination index of the route sequence: The internal step counter attribute within the targeted Delivery instance was incremented by 1 (attribute modification).
- If recordedWeight exceeds the maximum capacity weight of the associated Truck instance (Exception Paths):
  - If a destination replacement was executed, the target destination site association within the delivery route was updated to the newly selected alternative site (association modification).
  - If a truck replacement was executed, the associated Truck to the targeted Delivery instance was updated to the newly selected compatible truck (association modification).